AA-TV wiring v8: no wheels. 8x Deutsch DTP 4-pin (wet zone), cables epoxy-potted into dryboxes, 14 AWG marine outside / 20 AWG on boards.

BUOY (surface, dry)

5V_PI
BT2
USB-C power bank 5 V / 3 A
GND_PI

J1

GPIO12/PWM0	32	2	5V
GPIO13/PWM1	33	1	3V3
GPIO14/UART_TXD	8	27	ID_SD_I2C0/GPIO00
GPIO15/UART_RXD	10	28	ID_SC_I2C0/GPIO01
GPIO16/SPI1_CE2	36	3	SDA_I2C1/GPIO02
GPIO17/SPI1_CE1	11	5	SCL_I2C1/GPIO03
GPIO18/SPI1_CE0/PCM_CLK/PWM0	12	7	
GPIO19/SPI1_MISO/PCM_FS	35	29	GPCLK0/GPIO04
GPIO20/SPI1_MOSI/PCM_DIN/PWM1	38	31	GPCLK1/GPIO05
GPIO21/SPI1_SCLK/PCM_DOUT	40	33	GPCLK2/GPIO06
GPIO22/SDIO_CLK	15	26	CE1_SPI0/GPIO07
GPIO23/SDIO_CMD	16	24	CE0_SPI0/GPIO08
GPIO24/SDIO_DAT0	18	21	MISO_SPI0/GPIO09
GPIO25/SDIO_DAT1	22	19	MOSI_SPI0/GPIO10
GPIO26/SDIO_DAT2	37	23	SCLK_SPI0/GPIO11
GPIO27/SDIO_DAT3	13		
	6		GND

Raspberry Pi 4 (buoy)

Pi 4 talks to the Pico over USB serial (/dev/ttyACM0) and powers it from USB 5 V. Reset the Pico by power-cycling the Pi USB ports (uhubctl).

TETHER

W1

5V_PI	1	USB-A	
	2	micro-B	5
	3	+ USB data	8
GND_PI	4		
VBUS_PICO	5		
GND	8		

USB-A to micro-B, ~3 m

Vinyl air tube from the buoy; USB cable runs inside it. USB plugs are dry (buoy, Pico).

MAIN DRYBOX (upper, on hull)

PEN1
A1: air-tube fitting (USB inside)

BOARD A: Pico solder-board

GP12	R1 220	M4_PWM	
GP13	R2 220	M5_PWM	
GP14	R3 220	M6_PWM	
GP15	R4 220	M7_PWM	
GP16	R5 220	THR_PWM	
GP19	R6 220	PUMP_PWM	

Raspberry Pi Pico A1

	39	VSYS	
	40	VBUS	VBUS_PICO
	36	3V3	+3V3
RUN	30		
3V3_EN	37		
GPIO0	1		
GPIO1	2		
GPIO2	4		
GPIO3	5		
GPIO4	6		
GPIO5	7		
GPIO6	9		
GPIO7	10		
GPIO8	11		
GPIO9	12		
GPIO10	14		
GPIO11	15		
GPIO12	16	GP12	
GPIO13	17	GP13	
GPIO14	19	GP14	
GPIO15	20	GP15	
GPIO16	21		GP16
GPIO17	22		
GPIO18	24		
GPIO19	25		GP19
GPIO20	26		
GPIO21	27		
GPIO22	29		
ADC_VREF	35		
GPIO26_ADC0	31		LEAK_MAIN
GPIO27_ADC1	32		LEAK_REAR
GPIO28_ADC2	34		
AGND	33		GND
GND	3		

220 ohm in series on every PWM line that leaves the drybox.

TA2

1	+3V3	
2	GND	
3	LEAK_MAIN	

Main leak sensor

TA1

M4_PWM	1	
M5_PWM	2	
M6_PWM	3	
M7_PWM	4	
THR_PWM	5	
H1_SPARE6	6	
H1_SPARE7	7	
PUMP_PWM	8	
LEAK_REAR	9	
+3V3	10	
GND	11	
GND	12	

H1 out (14 AWG)

All Board A terminals: 5.08 mm pitch (0.2 in, fits the breadboard grid). TA1 pins 6-7: spare H1 conductors.

PEN2
A2: H1 cable, epoxy-potted

REAR DRYBOX (inside open rear endcap)

BOARD B: servo / signal board

TB1

M4_PWM	1	
M5_PWM	2	
M6_PWM	3	
M7_PWM	4	
THR_PWM	5	
H1_SPARE6	6	
H1_SPARE7	7	
PUMP_PWM	8	
LEAK_REAR	9	
+3V3	10	
GND	11	
GND	12	

H1 in (14 AWG)

TB7

1	SERVO_7V5	
2	GND	

7.5 V in (14 AWG)

SERVO_7V5
C1
1000uF 16V
GND

Board B:
- 7.5 V + GND rails: strip of 14 AWG soldered along strips
- Servo + / - jumpers: 14 AWG
- Signal jumpers: 20 AWG
- No 12 V on the breadboard
- TB1 pins 6-7: spare, unused

TB2

M4_PWM	1	
M5_PWM	2	
SERVO_7V5	3	
GND	4	

Rudder servos (to D4)

TB3

M6_PWM	1	
M7_PWM	2	
SERVO_7V5	3	
GND	4	

Elevator servos (to D5)

TB4

THR_PWM	1	
GND	2	

Thruster ESC signal (to D3)

TB5

PUMP_PWM	1	
GND	2	

Pump ESC signal (U3)

TB6

+3V3	1	
GND	2	
LEAK_REAR	3	

Rear leak sensor

Off-board, in rear drybox

BB1
VBAT
Bus bars (screw studs, ring terminals): in: D1 trunk, two 14 AWG per pole out: buck U1, pump ESC U3, D3 thruster ESC feed

+12 V bus bar (>=30 A)

BB2
GND

GND bus bar (>=30 A)

U1
VBAT
GND
VIN+
VIN-
BUCK
VOUT+
VOUT-
SERVO_7V5
GND

12 V -> 7.5 V buck (>=5 A)

U3
VBAT
GND
PUMP_PWM
BAT+
BAT-
BEC+
SIG
SIG_GND
M+
M-
ESC
PUMP_M+
PUMP_M-

Pump ESC (brushed, reversible)

Pump ESC sits dry, so any brushed ESC with reverse works. BEC wire cut. Each motor lead: dry splice to two 14 AWG wires for D2 pins 1-2 / 3-4.

PEN3
R2: signal cables potted (H1, 2x servo, THR sig)

PEN4
R3: power cables potted (trunk, thruster feed, pump)

WET ZONE (flooded hull, open rear endcap, splash areas): Deutsch DTP 4-pin connectors live here

20 A ATC, waterproof holder
F1
VBAT
BT1
3S LiPo (wet, no box)
SW1
GND
Kill switch, low side, >=30 A

Battery side: battery + -> F1 -> split to D1 pins 1, 2; battery - -> SW1 -> split to D1 pins 3, 4. Splits: solder + adhesive heat-shrink. Charge through a DTP04-4P -> XT60 adapter.

D1A D1B

VBAT	1	1	VBAT
VBAT	2	2	VBAT
GND	3	3	GND
GND	4	4	GND

D1 battery DTP06-4S DTP04-4P
D1: 2 contacts per pole (50 A).

D6A D6B

M4_PWM	1	1	M4_PWM
M5_PWM	2	2	M5_PWM
M6_PWM	3	3	M6_PWM
M7_PWM	4	4	M7_PWM

D6 H1 1-4 DTP06-4S DTP04-4P

D7A D7B

THR_PWM	1	1	THR_PWM
H1_SPARE6	2	2	H1_SPARE6
H1_SPARE7	3	3	H1_SPARE7
PUMP_PWM	4	4	PUMP_PWM

D7 H1 5-8 DTP06-4S DTP04-4P

D8A D8B

LEAK_REAR	1	1	LEAK_REAR
+3V3	2	2	+3V3
GND	3	3	GND
GND	4	4	GND

D8 H1 9-12 DTP06-4S DTP04-4P
H1 main <-> rear: 12 x 14 AWG over D6, D7, D8. Pins 8-7 spare.

D4A D4B

M4_PWM	1	1	M4_PWM
M5_PWM	2	2	M5_PWM
SERVO_7V5	3	3	SERVO_7V5
GND	4	4	GND

D4 rudder DTP06-4S DTP04-4P

D5A D5B

M6_PWM	1	1	M6_PWM
M7_PWM	2	2	M7_PWM
SERVO_7V5	3	3	SERVO_7V5
GND	4	4	GND

D5 elevator DTP06-4S DTP04-4P
Servo side: each lead spliced to 14 AWG; the two + leads share pin 3, the two GND share pin 4.

M4_PWM, M5_PWM, SERVO_7V5, GND -> Rudder servo
M6_PWM, M7_PWM, SERVO_7V5, GND -> Elevator servo

D3A D3B

VBAT	1	1	VBAT
GND	2	2	GND
THR_PWM	3	3	THR_PWM
GND	4	4	GND

D3 thruster DTP06-4S DTP04-4P
D3: 1 +12 V, 2 GND, 3 PWM (yellow), 4 signal GND (white). All 14 AWG.

U2
VBAT
GND
THR_PWM
GND
BAT+
BAT-
SIG
SIG_GND
BLDC ESC
U
V
W
MT1
BLDC
TD1.2 thruster

TD1.2 ESC (waterproof, wet)
Factory thruster cable stays on the ESC.

D2A D2B

PUMP_M+	1	1	PUMP_M+
PUMP_M+	2	2	PUMP_M+
PUMP_M-	3	3	PUMP_M-
PUMP_M-	4	4	PUMP_M-

D2 pump DTP06-4S DTP04-4P
D2: 2 contacts per pump lead.

PUMP_M+
M3
Ballast pump, 12 V brushed (TBD, <=5 A)
PUMP_M-

Deutsch DTP: size 12 contacts, 12-14 AWG, 25 A each. Every cavity is filled (no seal plugs needed). Source side (battery / drybox) gets the DTP06 plug with socket contacts.

Wire: 14 AWG marine for everything outside the boxes, every DTP contact and all power; 20 AWG hookup inside the boxes.
Only connectors: 8x Deutsch DTP 4-pin in the wet zone + 5.08 mm screw terminals / bus bars inside the boxes.
Two solder-boards (Board A main drybox, Board B rear drybox). Cables epoxy-potted through drybox walls.

PIE AA-TV
Sheet: /
File: AATV.kicad_sch

Title: AA-TV electrical v8: no wheels, Deutsch DTP 4-pin connectors

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